ChE 311

HW #12

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10

I. 1) Laminar flow

$$V_z = V_{z,max} \cdot \left(1 - \left(\frac{r}{R}\right)^2\right)$$

$$\langle V_z \rangle = \frac{\int_0^R V_{z,max} \cdot 2\pi r \left(1 - \left(\frac{r}{R}\right)^2\right) dr}{\pi R^2} = \frac{2 V_{z,max} \int_0^R r - \frac{r^3}{R^2}\, dr}{R^2} = \frac{2 V_{z,max} \left(\frac{R^2}{2} - \frac{R^4}{4R^2}\right) dr}{R^2}$$

$$= \frac{V_{z,max}}{2}$$

$$\langle V_z^2 \rangle = \frac{\int_0^R V_{z,max}^2 \cdot 2\pi r \left(1 - \frac{r^2}{R^2}\right)^2 dr}{\pi R^2} = \frac{2 V_{z,max}^2 \int_0^R r - \frac{2r^3}{R^2} + \frac{r^5}{R^4}\, dr}{R^2} = \frac{2 V_{z,max}^2 \left(\frac{R^2}{2} - \frac{2R^4}{4R^2} + \frac{R^6}{6R^4}\right)}{R^2}$$

$$= \frac{V_{z,max}^2}{3}$$

$$\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{\frac{V_{z,max}^2}{3}}{\frac{V_{z,max}^2}{4}} = \frac{4}{3}$$

2) turbulent flow:

$$\bar{V}_z = \bar{V}_{z,max} \left(1 - \frac{r}{R}\right)^{\frac{1}{7}}$$

wolfram

$$\langle \bar{V}_z \rangle = \frac{\int_0^R \bar{V}_{z,max} \cdot 2\pi r \left(1 - \frac{r}{R}\right)^{\frac{1}{7}} dr}{\pi R^2} = \frac{2 \bar{V}_{z,max} \left(\int_0^R r \left(1 - \frac{r}{R}\right)^{\frac{1}{7}} dr\right)}{R^2} = \frac{2 \bar{V}_{z,max} \frac{49R^2}{120}}{R^2}$$

$$= \frac{\bar{V}_{z,max} \cdot 49}{60}$$

wolfram

$$\langle \bar{V}_z^2 \rangle = \frac{\int_0^R \bar{V}_{z,max}^2 \cdot 2\pi r \left(1 - \frac{r}{R}\right)^{\frac{2}{7}} dr}{\pi R^2} = \frac{2 \bar{V}_{z,max}^2 \left(\int_0^R r \left(1 - \frac{r}{R}\right)^{\frac{2}{7}} dr\right)}{R^2} = \frac{2 \bar{V}_{z,max}^2 \frac{49R^2}{144}}{R^2}$$

$$= \frac{\bar{V}_{z,max}^2 \cdot 49}{72}$$

$$\frac{\langle \bar{V}_z^2 \rangle}{\langle \bar{V}_z \rangle^2} = \frac{\frac{\cancel{\bar{V}_{z,max}^2} \cdot \cancel{49}}{72}}{\frac{\cancel{\bar{V}_{z,max}^2} \cdot 49^2}{3600}} = \frac{3600}{3528} = \frac{50}{49}$$

2. $\bar{v}_z = \bar{v}_{z,max}\left(1-\frac{r}{R}\right)^{\frac{1}{n}}$

$$\langle \bar{v}_z \rangle = \frac{\int_0^R \bar{v}_z \cdot 2\pi r\, dr}{\pi R^2} = \frac{2\cdot \bar{v}_{z,max}\left(\int_0^R r\left(1-\frac{r}{R}\right)^{\frac{1}{n}} dr\right)}{R^2} \xrightarrow{\text{wolfram}} = \frac{2\bar{v}_{z,max}\left(\frac{n^2R^2}{2n^2+3n+1}\right)}{R^2}$$

$$= \frac{2\bar{v}_{z,max}\, n^2}{2n^2+3n+1}$$

$$\frac{\langle \bar{v}_z \rangle}{\bar{v}_{z,max}} = \frac{2\bar{v}_{z,\cancel{max}}\, n^2}{(2n^2+3n+1)\cdot \bar{v}_{z,\cancel{max}}} = \frac{2n^2}{2n^2+3n+1} = \frac{2n^2}{(2n+1)(n+1)}$$